

Bevacizumab plus erlotinib versus erlotinib alone in Japanese patients with advanced, metastatic, *EGFR*-mutant non-small-cell lung cancer (NEJ026): overall survival analysis of an open-label, randomised, multicentre, phase 3 trial



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Summary

Background Bevacizumab is a promising candidate for combination treatment with epidermal growth factor receptor tyrosine-kinase inhibitors (eg, erlotinib), which could improve outcomes for patients with metastatic *EGFR*-mutant non-small-cell lung cancer (NSCLC). We have previously shown in NEJ026, a phase 3 trial, that the combination of bevacizumab plus erlotinib significantly prolonged progression-free survival compared with erlotinib alone in these patients. In further analyses, we aimed to examine the effects of bevacizumab–erlotinib on overall survival, time from enrolment to progressive disease during second-line treatment or death, and quality of life.

Methods This open-label, randomised, multicentre, phase 3 trial (NEJ026) was done in 69 hospitals and medical, community-based centres across Japan. Eligible patients had stage IIIB, stage IV, or postoperative recurrent, *EGFR*-mutant (exon 19 deletion or exon 21 Leu858Arg point mutation) NSCLC, had not previously received systemic chemotherapy, and were randomly assigned (1:1) by a computer-generated randomisation sequence and minimisation to receive either 150 mg oral erlotinib once daily plus 15 mg/kg intravenous bevacizumab once every 21 days, or 150 mg oral erlotinib once daily, until disease progression or intolerable toxicity. Randomisation was stratified according to sex, smoking status, *EGFR* mutation subtype, and clinical disease stage. All participants, investigators, and study personnel (including those assessing outcomes) were unmasked to treatment allocation. We report the secondary outcomes of overall survival and quality of life (the period from enrolment to confirmation of a minimally important difference on the European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire [EORTC QLQ-C30]), and the exploratory outcome of time from enrolment to progressive disease during second-line treatment or death. Overall survival and the exploratory outcome were analysed in the modified intention-to-treat population, which comprised all randomly assigned patients who received at least one dose of the study drug and had response evaluations. Quality of life was analysed in patients in the modified intention-to-treat population who had completed the quality of life questionnaires. The trial is registered with the University Hospital Medical Information Network Clinical Trials Registry, UMIN000017069, and the Japan Registry of Clinical Trials, jRCTs031180056, and is currently closed.

Findings Between June 3, 2015, and Aug 31, 2016, 228 patients were enrolled. 112 patients who received bevacizumab–erlotinib and 112 who received erlotinib only were included in the modified intention-to-treat population. At data cutoff (Nov 30, 2019) and a median follow-up of 39·2 months (IQR 23·9–43·5), the median overall survival was 50·7 months (95% CI 37·3–not estimable [NE]) in the bevacizumab–erlotinib group and 46·2 months (38·2–NE) in the erlotinib-only group (hazard ratio [HR] 1·007, 95% CI 0·681–1·490; $p=0·97$). In analysis of the exploratory outcome, after a median follow-up of 23·9 months (IQR 14·2–39·1), the median time from enrolment to progressive disease during second-line treatment or death was 28·6 months (95% CI 22·1–35·9) in the bevacizumab–erlotinib group and 24·3 months (20·4–29·1) in the erlotinib-only group (HR 0·773, 95% CI 0·562–1·065). The median time between enrolment and confirmation of a minimally important difference on the EORTC QLQ-C30 was 6·0 months (95% CI 5·2–11·3) in the bevacizumab–erlotinib group and 8·3 months (5·7–13·9) in the erlotinib-only group ($p=0·47$).

Interpretation The addition of bevacizumab to erlotinib did not prolong survival in patients with metastatic *EGFR*-mutant NSCLC, but both treatment groups had relatively long survival durations. Why the addition of bevacizumab to erlotinib did not affect overall survival is unclear, but it is possible that the beneficial effects of combination therapy were not seen because overall survival was influenced by treatment regimens used after disease progression.

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Research in context

Evidence before this study

Epidermal growth factor receptor (EGFR) tyrosine-kinase inhibitors (EGFR TKIs) have been established as the standard first-line treatment for patients with EGFR-mutant non-small-cell lung cancer (NSCLC). Despite durable responses, median progression-free survival is around 1 year with first-generation EGFR TKIs (gefitinib and erlotinib) as a result of acquired therapeutic resistance. Combination treatments with EGFR TKIs are attractive options to improve outcomes for these patients. We searched PubMed for articles published in English between Jan 1, 2010, and Dec 31, 2015, using the search string “erlotinib”, “bevacizumab”, “NSCLC”, and “phase 2 or phase 3”. We also searched ClinicalTrials.gov for clinical trials in English published between Jan 1, 2010, and Dec 31, 2020, using the search string “erlotinib”, “bevacizumab”, and “NSCLC”, filtering for phase 3 studies that were recruiting or active. We identified two studies that assessed the activity of bevacizumab plus erlotinib for patients with NSCLC: the BeTa study and the JO25567 study. The BeTa study was a phase 3 trial that compared bevacizumab–erlotinib with erlotinib alone in patients with advanced NSCLC after standard first-line chemotherapy had failed. In the BeTa study, median progression-free survival for patients with EGFR-mutant NSCLC treated with bevacizumab–erlotinib (17.1 months) was longer than for patients treated with erlotinib only (9.7 months; hazard ratio [HR] 0.52, 95% CI 0.19–1.38), although this analysis was post-hoc and the sample size was small. The JO25567 study was a randomised, multicentre, phase 2 trial that compared bevacizumab–erlotinib with erlotinib alone as first-line treatments in 154 Japanese patients with advanced EGFR-mutant NSCLC. In the JO25567 study, bevacizumab–erlotinib prolonged median progression-free survival (16.0 months, 95% CI 13.9–18.1) compared with erlotinib alone (9.7 months, 5.7–11.1; HR 0.54, 95% CI 0.36–0.79).

Added value of this study

To our knowledge, our study is the only phase 3 study to compare bevacizumab–erlotinib with erlotinib alone as first-line treatments in patients with advanced EGFR-mutant NSCLC. The interim analysis of NEJ026, with a data cutoff date of Sept 21, 2017, has already been reported: it showed that median

progression-free survival was significantly longer in the bevacizumab–erlotinib group (16.9 months, 95% CI 14.2–21.0) than in the erlotinib-only group (13.3 months, 11.1–15.3; HR 0.605, 95% CI 0.417–0.877; $p=0.016$). However, we now report that the median overall survival did not differ between the bevacizumab–erlotinib group (50.7 months, 95% CI 37.3–not estimable [NE]) and the erlotinib-only group (46.2 months, 38.2–NE; HR 1.007, 95% CI 0.681–1.490; $p=0.97$).

Implications of all the available evidence

No study has shown the superiority of bevacizumab–erlotinib over erlotinib alone with regard to overall survival. Likewise, in the JO25567 study, there was no difference in median overall survival between the bevacizumab–erlotinib group (47.0 months) and the erlotinib-only group (47.4 months; HR 0.81, 95% CI 0.53–1.23; $p=0.3267$). There might be several reasons why bevacizumab did not impact overall survival. Therefore, when considering overall survival, whether dual inhibition of EGFR and vascular endothelial growth factor pathways is an optimal strategy in the treatment of patients with EGFR-mutant NSCLC is controversial. The RELAY study was a worldwide, double-blind, placebo-controlled, phase 3 study of patients with untreated, EGFR-mutant, advanced NSCLC done in 13 countries (including Japan). The trial found that erlotinib plus ramucirumab, a human monoclonal IgG1 VEGFR-2 antagonist, resulted in longer progression-free survival compared with erlotinib plus placebo (19.4 months [95% CI 15.4–21.6] vs 12.4 months [11.0–13.5]; HR 0.59, 95% CI 0.46–0.76). The overall survival data were immature at the time of data cutoff. Another phase 3 study done in Chinese patients (the CTONG1509 study) found that median progression-free survival after bevacizumab–erlotinib treatment (18.0 months, 95% CI 15.2–20.7) was significantly longer than that after erlotinib treatment (11.3 months, 9.8–13.8; HR 0.55, 95% CI 0.41–0.75; $p<0.001$). However, the overall survival data have not yet been reported. Moreover, one phase 3 study comparing bevacizumab–erlotinib with erlotinib alone in patients with EGFR-mutant NSCLC is in progress (BEVERLY; NCT02633189). The overall survival results from these studies will help to establish whether this treatment strategy is optimal.

Introduction

Epidermal growth factor receptor (EGFR) tyrosine-kinase inhibitors (TKIs) are the standard-of-care first-line treatment for patients with advanced, metastatic non-small-cell lung cancer (NSCLC) positive for mutations in EGFR. Despite durable responses, median progression-free survival is around 1 year with first-generation EGFR TKIs (gefitinib and erlotinib) as a result of acquired therapeutic resistance.^{1–3}

Osimertinib, a third-generation EGFR TKI, elicited longer progression-free survival and overall survival than did the first-generation EGFR TKIs, erlotinib and

gefitinib, in the FLAURA study.^{4,5} Thus, osimertinib has been adopted as the standard first-line treatment for patients with advanced, EGFR-mutant, metastatic NSCLC. However, there are no standard targeted treatment options after disease progression on osimertinib; patients are usually treated with cytotoxic chemotherapy and immunotherapy, which provide little benefit.

Combination treatments with EGFR TKIs are attractive options to improve outcomes for these patients. Bevacizumab is an anti-angiogenic monoclonal antibody that targets the vascular endothelial growth factor (VEGF) signalling pathway and is a promising candidate for

combination treatments with EGFR TKIs. Bevacizumab inhibits tumour angiogenesis, improves the delivery of EGFR TKIs through vascular normalisation, alleviates immunosuppression, and promotes efficient tumour infiltration by effector immune cells.⁶

The first phase 3 study comparing the activity of bevacizumab–erlotinib with the activity of erlotinib alone in patients with NSCLC (not stratified by *EGFR* mutation status) was the BeTa study.⁷ Although no substantial differences in overall survival were identified between the groups, subgroup analyses suggested that the combination therapy provided an overall survival benefit in patients with *EGFR*-mutant NSCLC.

JO25567 was a phase 2 study⁸ that compared bevacizumab plus erlotinib treatment with erlotinib alone in 154 Japanese patients with advanced, *EGFR*-mutant, metastatic NSCLC; its primary endpoint was progression-free survival and secondary endpoints were overall survival, tumour response, quality of life, symptom improvement, and safety. In the JO25567 study, bevacizumab–erlotinib elicited a significant improvement compared with erlotinib alone in terms of median progression-free survival (16.0 months [95% CI 13.9–18.1] vs 9.7 months [5.7–11.1]).⁸ However, the study was inadequately powered to assess overall survival and did not reveal any improvement in this measure with the addition of bevacizumab.⁹

In view of the potential promise of combination treatments with EGFR TKIs and the results for progression-free survival obtained in the JO25567 study, we decided to examine the benefits of combination therapy with bevacizumab–erlotinib in a phase 3 trial. At the time of this decision, osimertinib had not yet been adopted as the first-line treatment for patients with *EGFR*-mutant, metastatic NSCLC. Therefore, we aimed to compare bevacizumab–erlotinib versus erlotinib alone in the treatment of advanced *EGFR*-mutant metastatic NSCLC in the phase 3 trial NEJ026. The interim analysis of NEJ026, with a data cutoff date of Sept 21, 2017, has already been reported.¹⁰ This analysis showed that the primary endpoint of progression-free survival was significantly longer in the bevacizumab–erlotinib group (median 16.9 months, 95% CI 14.2–21.0) than in the erlotinib group (13.3 months, 11.1–15.3; hazard ratio [HR] 0.605, 95% CI 0.417–0.877; $p=0.016$).¹⁰ Furthermore, the results of the NEJ026 biomarker study revealed that monitoring plasma *EGFR* mutation status provided important information that could facilitate assessment of the efficacy of the current treatment (bevacizumab–erlotinib and erlotinib alone) and represented a candidate prognostic test.¹¹

Here, we report the final overall survival analysis, data on time from enrolment to progressive disease during second-line treatment or death, and data for quality of life from the NEJ026 study.

Methods

Study design and participants

NEJ026 was an open-label, randomised, multicentre, phase 3 trial done in 69 hospitals and medical community-based centres across Japan. The investigators recruited eligible patients at each site. Eligible patients had histologically or cytologically confirmed advanced, metastatic, stage IIIB, stage IV (according to the 7th edition of the *General Rule for Clinical and Pathological Record of Lung Cancer*),¹² or postoperative recurrent non-squamous NSCLC, with activating *EGFR* mutations (either an exon 19 deletion or an exon 21 Leu858Arg point mutation). Other eligibility criteria included being 20 years or older when giving informed consent and having an Eastern Cooperative Oncology Group (ECOG) performance status of 0–2, one or more measurable lesions (based on Response Evaluation Criteria in Solid Tumours [RECIST], version 1.1), and a life expectancy of at least 3 months. Patients were excluded if any of the following applied: detection of an exon 20 Thr790Met mutation in *EGFR* in their tumour samples (but not if detected in plasma samples); brain metastases requiring anti-oedema drugs, such as corticosteroids; previous malignancies with a disease-free period of less than 5 years; massive effusion requiring drainage; a history of, or current, haemoptysis; a history of, or current, serious complications (eg, interstitial lung diseases); or previous treatment with systemic chemotherapy. All eligibility criteria and details of the laboratory tests required for assessing eligibility are provided in the appendix (pp 53–56).

See Online for appendix

The study was done in accordance with the Declaration of Helsinki and Good Clinical Practice Guidelines. The study protocol was reviewed and approved by the ethics committee of the non-profit organisation MINS Institutional Review Board and by the local institutional review boards of the participating institutions, and can be found in the appendix (pp 26–92). Written informed consent was obtained from all enrolled participants.

Randomisation and masking

Patients were centrally randomly assigned (1:1) by a computer-generated randomisation sequence and minimisation at a NEJ026 data centre (AC Medical, Tokyo, Japan) to receive either bevacizumab–erlotinib or erlotinib alone by use of patient enrolment sheets provided by the investigators. Randomisation was stratified according to sex (female or male), smoking status (non-smoker, former light smoker, or other smoker), *EGFR* mutation subtype (exon 19 deletion or exon 21 Leu858Arg point mutation), and clinical stage (stage IIIB, stage IV, or postoperative recurrence). Former light smokers were defined as individuals who had smoked less than 10 pack-years (pack-years was defined as duration [years]×quantity [sticks per day] divided by 20) in their lifetime and had stopped smoking more than 15 years before the study. The

NEJ026 data centre generated the randomisation sequence, confirmed participant eligibility, assigned the eligible patients to trial groups, and notified investigators of treatment allocation for each patient. All patients, investigators, and study personnel (including those assessing outcomes) were unmasked to treatment allocation.

Procedures

During screening, investigators assessed patients' bodyweight, ECOG performance status, and vital signs, and performed laboratory tests (eg, complete blood count, blood biochemistry, coagulation, urinalysis, and arterial blood oxygen saturation). Additionally, patients had imaging tests (eg, chest x-rays, chest CTs, brain MRIs, and PET scans) to assess tumour spread and to confirm that no new lesions were present during screening. *EGFR* mutations were screened by high-sensitivity *EGFR* gene mutation tests covered by health insurance (with a peptide nucleic acid-locked nucleic acid PCR clamp method [LSI Medience Corporation, Tokyo, Japan] or other test methods) using tumour tissue or tumour cells.

Patients assigned to the bevacizumab–erlotinib group received 150 mg of oral erlotinib in tablet form once daily and 15 mg/kg of intravenous bevacizumab once every 21 days. Patients assigned to the erlotinib-only group received 150 mg of oral erlotinib once daily. One treatment cycle was defined as 21 days, and patients remained on treatment until disease progression or intolerable toxicity. Physical examinations (bodyweight measurements, ECOG performance status, and vital signs), laboratory tests (complete blood count, blood biochemistry, coagulation, urinalysis, and partial pressure of oxygen in arterial blood or arterial blood oxygen saturation), and chest x-rays were done at baseline and during every cycle. Two steps of dose reductions of erlotinib were permitted (to 100 mg/day and 50 mg/day) if a dose reduction criterion was fulfilled (appendix pp 30–32). Dose reductions of bevacizumab were not permitted except when dose adjustment was needed because of a change in bodyweight. Erlotinib administration could be delayed for up to 3 weeks and bevacizumab administration could be delayed for up to 6 weeks (appendix p 31). In the bevacizumab–erlotinib group, if either drug was delayed or discontinued, the other drug could be continued.

We defined the study period as consisting of two parts: the period between patient enrolment and progressive disease after first-line treatment (period 1) and the observation period between progressive disease after first-line treatment and progressive disease during second-line treatment (period 2). The observation period was terminated unless another treatment was started within 42 days of the end of period 1. Bevacizumab–erlotinib or erlotinib alone, used as the next study treatment after period 1, were not regarded as second-line treatments but were permitted. In the NEJ026 study,

platinum-based chemotherapy (intravenous carboplatin [area under the curve 5–6 every 21 days] or intravenous cisplatin [75 mg/m² every 21 days]) plus intravenous pemetrexed (500 mg/m² every 21 days) was recommended for the bevacizumab–erlotinib group and platinum-based chemotherapy (carboplatin or cisplatin) plus intravenous pemetrexed and intravenous bevacizumab (15 mg/kg every 21 days) was recommended for the erlotinib-only group as second-line treatments, according to the protocol, but other second-line treatments were permissible (appendix pp 32–33).

Tumour assessments were done by investigators within the 28 days before enrolment (at baseline), every 6 weeks until 18 months after treatment initiation, and every 12 weeks thereafter until disease progression according to RECIST, version 1.1 (the end of period 1). During the observation period (period 2), continuation of tumour assessments at the same follow-up schedule was recommended if possible. Unless there was any special reason, target lesions and non-target lesions were monitored by the same test method (CT and MRI) and under the same conditions (by use of the same slice thickness and contrast agent) as the evaluation method used during screening. Similarly, we utilised the same evaluation method used during screening to confirm that no new lesions were present.

Quality of life was assessed before treatment initiation (at baseline) and every 12 weeks after treatment initiation until the end of period 1; continuation of these quality-of-life assessments during the observation period (period 2) was done every year after period 1 until disease progression, if possible. The European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire (EORTC QLQ)-C30 (version 3.0)¹³ or the EORTC QLQ-LC13, which specifically includes a 13-item lung cancer-specific section, was used to obtain measures of quality of life.¹⁴

Outcomes

The primary endpoint was centrally assessed progression-free survival, defined as the time from randomisation to disease progression with first-line treatment or death, and has already been reported.¹⁰ Secondary endpoints reported herein were overall survival (defined as the time from randomisation to death due to any cause) and quality of life (based on the EORTC QLQ-C30; although we collected EORTC QLQ-LC13 data, we are yet to analyse these). Other secondary endpoints were objective response rate (defined as the proportion of patients with a complete or partial response), disease control rate (defined as the proportion of patients with a complete response, a partial response, or stable disease), duration of response (defined as the time from confirmation of a complete or partial response to disease progression), and safety, which were previously reported.¹⁰ Exploratory endpoints were time from enrolment to progressive disease during second-line treatment or death (period 2;

reported herein), biomarker analyses of the relationship between changes in *EGFR* mutation status and therapeutic effect (reported separately),¹¹ and a pooled analysis (with the JO25567 study)⁸ of overall survival, which we hope to report at a later date.

Statistical analysis

On the basis of results from the JO25567 study,⁸ we expected a median progression-free survival of 10 months in the erlotinib group and 16 months in the bevacizumab-erlotinib group. With this assumption, the number of progression-free survival events required to detect a HR of 0.63 with a two-sided significance level of 5% and a power of 80% was 147. Assuming an enrolment period of 18 months and an observation period of 18 months, the required number of patients was 214 (107 in each group). The interim analysis was planned when 67% of the required progression-free survival events were observed. An independent data monitoring committee oversaw the study.

Efficacy endpoints (eg, overall survival and time from enrolment to progressive disease during second-line treatment or death) were analysed in the modified intention-to-treat population, which comprised all randomly assigned patients who received at least one dose of the study drug and had response evaluations. Quality of life was analysed in patients in the modified intention-to-treat population who had completed the quality of life questionnaires.

Kaplan–Meier curves were drawn for time from enrolment to progressive disease during second-line treatment or death and overall survival and compared by use of a stratified log-rank test with a significance level of 0.05. HRs and associated 95% CIs were calculated via a Cox proportional hazards analysis. Using the same statistical methods, we did prespecified subgroup analyses of overall survival, stratifying patients on the basis of *EGFR* mutation status, the presence or absence of brain metastases, and other prespecified patient characteristics (ie, sex, age, ECOG performance status, smoking status, disease stage, and the presence or absence of pleural involvement; appendix p 4). In a post-hoc subgroup analysis, we examined overall survival in terms of whether patients were or were not treated with osimertinib in the second line, or in the second line or later.

The primary analysis in the quality-of-life assessment was the period from enrolment to confirmation of a minimally important difference, defined as a decline of 1 or more from the total combined score of items 29 and 30 on the EORTC QLQ-C30. Item 29 indicates overall health during the past week and item 30 indicates overall quality of life during the past week. The scores range from 1 (very poor) to 7 (excellent). Kaplan–Meier curves and the log-rank test were used to compare the period from enrolment to confirmation of a minimally important difference between treatment groups. The secondary analyses of quality of life were

comparisons of the periods between enrolment and confirmation of a decline of 2 or more, and between enrolment and a confirmation of a decline of 3 or more, in the total combined score of items 29 and 30 of the EORTC QLQ-C30. In post-hoc analyses, we examined changes in patient-reported outcomes with time: global health status (quality of life; EORTC QLQ-C30 questions 29 and 30), functional scales (physical in EORTC QLQ-C30 questions 1–5; role in questions 6 and 7; cognitive in questions 20 and 25; emotional in questions 21–24; and social in questions 26 and 27), and three prespecified patient-reported outcomes (fatigue in EORTC QLQ-C30 questions 10, 12, and 18; pain in questions 9 and 19; and nausea or vomiting in questions 14 and 15). For each scale or item, a linear transformation was applied to standardise the raw score to a range of 0–100 (depending on the item, high scores represented a high or healthy level of functioning or a high or severe level of symptomatology).

All reported p values are two-sided. Statistical analyses were done with SAS for Windows, version 9.4. The trial is registered with the University Hospital Medical Information Network Clinical Trials Registry, number UMIN000017069, and the Japan Registry of Clinical Trials, number jRCTs031180056.

Role of the funding source

The funder of the study approved the study design and was involved in the provision of information on the drug, but had no role in data collection, data analysis, data interpretation, or writing of the report.

Results

Between June 3, 2015, and Aug 31, 2016, 228 patients were enrolled and randomly assigned to receive either bevacizumab-erlotinib (n=114) or erlotinib alone (n=114; figure 1). Of the 228 patients who were randomly assigned, two withdrew from the study before treatment initiation and two were randomly assigned in error and were also excluded, leaving 224 patients in the modified intention-to-treat population (figure 1). Of these 224 patients, 12 were excluded from the quality-of-life analysis because of an absence of data (figure 1).

The patients' baseline characteristics are summarised in the table. Although adding bevacizumab was permitted in the erlotinib-only group after the interim analysis, no patients received the additional drug. In the bevacizumab-erlotinib group, the median duration of erlotinib treatment was 405 days (IQR 204–499) and the median duration of bevacizumab treatment was 350 days (146–469). In the erlotinib-only group, the median duration of erlotinib treatment was 364 days (IQR 168–479). The dose of erlotinib was reduced in 15 (13%) of 112 participants in the bevacizumab-erlotinib group and in 17 (15%) of 112 participants in the erlotinib-only group. In the bevacizumab-erlotinib group, erlotinib administration was delayed in 51 (46%)

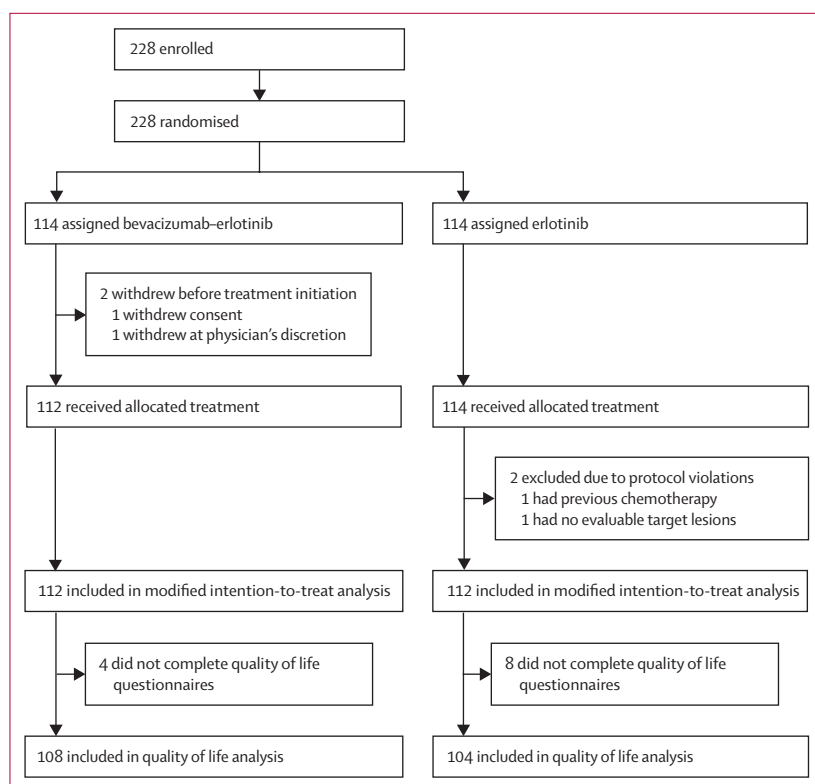


Figure 1: Trial profile

patients and bevacizumab administration was delayed in 65 (58%) patients. In the erlotinib-only group, erlotinib administration was delayed in 39 (35%) patients.

As previously reported,¹⁰ at the data cutoff point of the preplanned interim analysis (Sept 21, 2017), 117 patients had a progression-free survival event, which represented 80% of the 147 prespecified final number of events required. The independent data monitoring committee held a meeting on Jan 23, 2018, and recommended early termination of the study on the basis of results from the interim analysis. However, to obtain data for endpoints other than progression-free survival, the study was continued.

At the last data cutoff point (Nov 30, 2019), one (1%) patient in the bevacizumab-erlotinib group and four (4%) patients in the erlotinib-only group were still on the protocol treatment. At a median follow-up of 39.2 months (IQR 23.9–43.5), in the analysis of overall survival, 51 (46%) deaths had occurred in the bevacizumab-erlotinib group and 50 (45%) deaths had occurred in the erlotinib-only group. The median overall survival was 50.7 months (95% CI 37.3–not estimable [NE]) in the bevacizumab-erlotinib group and 46.2 months (38.2–NE) in the erlotinib-only group (HR 1.007, 95% CI 0.681–1.490; $p=0.97$; figure 2). Analysis of overall survival stratified into patient subgroups, including by *EGFR* mutation status and the presence or absence of

	Bevacizumab-erlotinib (n=112)	Erlotinib (n=112)
Age		
Median, years	67 (61–73)	68 (62–73)
<75 years	92 (82%)	89 (79%)
≥75 years	20 (18%)	23 (21%)
Sex		
Female	71 (63%)	73 (65%)
Male	41 (37%)	39 (35%)
Smoking status		
Never smoker	65 (58%)	64 (57%)
Former light smoker*	6 (5%)	7 (6%)
Other smoker	41 (37%)	41 (37%)
ECOG performance status		
0	64 (57%)	68 (61%)
1	48 (43%)	42 (38%)
2	0	2 (2%)
Pathology		
Adenocarcinoma	110 (98%)	112 (100%)
Large cell carcinoma	1 (1%)	0
Other	1 (1%)	0
EGFR mutation type		
Exon 19 deletion	56 (50%)	55 (49%)
Exon 21 Leu858Arg point mutation	56 (50%)	57 (51%)
Stage at screening		
IIIB	8 (7%)	8 (7%)
IV	82 (73%)	84 (75%)
Postoperative recurrence	22 (20%)	20 (18%)
Pleural involvement		
Yes	45 (40%)	46 (41%)
No	67 (60%)	66 (59%)
Brain metastases		
Yes	36 (32%)	36 (32%)
No	76 (68%)	76 (68%)

Data are median (IQR) or n (%). Most of these data have been previously reported.¹⁰ ECOG=Eastern Cooperative Oncology Group. *Former light smokers were defined as individuals who had smoked less than 10 pack-years in their lifetime and had stopped smoking more than 15 years before the study.

Table: Baseline patient characteristics in the modified intention-to-treat population

brain metastases, showed no difference between the treatment groups, but should be interpreted with caution as these analyses did not have adequate power (appendix p 4). Kaplan–Meier curves for overall survival stratified by *EGFR* mutation type, pleural involvement, and the presence of brain metastases, can be found in the appendix (pp 5–7).

Details of post-protocol treatments in each group can be found in the appendix (pp 2–3, 8–9). The cumulative total regimen numbers after first-line treatment were 187 in the bevacizumab-erlotinib group and 255 in the erlotinib-only group. Specifically, 57 (51%) patients in the bevacizumab-erlotinib group and 72 (64%) patients in

the erlotinib-only group received cytotoxic chemotherapies, 29 (26%) patients in the bevacizumab-erlotinib group and 43 (38%) patients in the erlotinib-only group received EGFR TKIs other than osimertinib, and 13 (12%) patients in the bevacizumab-erlotinib group and 25 (22%) patients in the erlotinib-only group received immunotherapy (appendix p 3). 30 (27%) patients in the bevacizumab-erlotinib group and 46 (41%) patients in the erlotinib-only group received three or more regimens after the first-line protocol treatment (ie, fourth-line; appendix p 22). According to the protocol recommendation, 33 (29%) patients in the bevacizumab-erlotinib group received platinum plus pemetrexed and 32 (29%) patients in the erlotinib-only group received platinum plus pemetrexed plus bevacizumab as a second-line treatment. Five patients in the erlotinib-bevacizumab group and six patients in the erlotinib-only group transitioned to erlotinib alone or erlotinib-bevacizumab after progressive disease, although these were not considered second-line treatments. 31 (28%) patients in the bevacizumab-erlotinib group and 29 (26%) patients in the erlotinib-only group had Thr790Met *EGFR* mutations detected at disease progression during first-line treatment and received osimertinib in the second line. In the bevacizumab-erlotinib group, the median overall survival was 50.7 months (95% CI 37.3–50.7) for patients treated with second-line osimertinib and 37.6 months (27.8–NE) for patients not treated with second-line osimertinib (HR 0.627, 95% CI 0.327–1.202; appendix p 10). In the erlotinib-only group, the median overall survival was not reached in patients treated with second-line osimertinib and was 40.1 months (95% CI 26.7–NE) in patients not treated with second-line osimertinib (HR 0.647, 95% CI 0.328–1.278; appendix p 10). Osimertinib was administered in the second line, third line, and later lines in both groups. Considering all treatment courses, 48 (43%) patients in the bevacizumab-erlotinib group and 50 (45%) patients in the erlotinib-only group had Thr790Met mutations detected and received osimertinib. In the bevacizumab-erlotinib group, the median overall survival was 50.7 months (95% CI 37.3–NE) for patients treated with osimertinib in the second line or later and 33.5 months (24.0–NE) for patients not treated with osimertinib in the second line or later (HR 0.610, 95% CI 0.333–1.119; appendix p 11). In the erlotinib-only group, the median overall survival was 46.2 months (95% CI 38.2–NE) for patients treated with osimertinib in the second line or later and 30.6 months (23.5–NE) for patients not treated with osimertinib in the second line or later (HR 0.606, 95% CI 0.337–1.089; appendix p 11).

In our quality-of-life analysis, the median time between enrolment and confirmation of a minimally important difference did not differ between the treatment groups ($p=0.47$; figure 3). The median time between enrolment and confirmation of a decline of 2 or more in the total combined score of items 29 and 30 of the EORTC

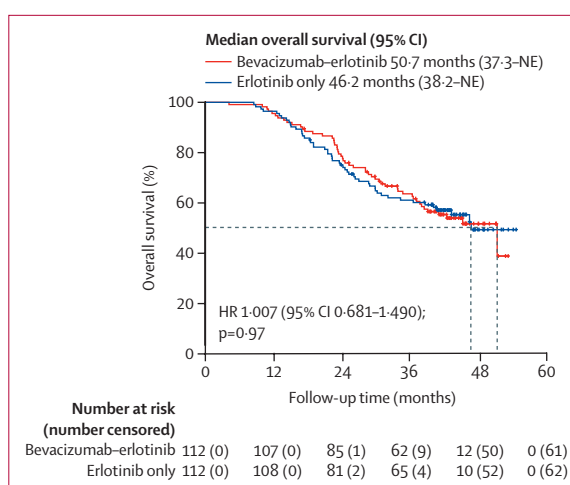


Figure 2: Kaplan-Meier curves for overall survival in the modified intention-to-treat population

The tick marks indicate censored observations. The two-sided p value was calculated by use of a log-rank test. HR=hazard ratio. NE=not estimable.

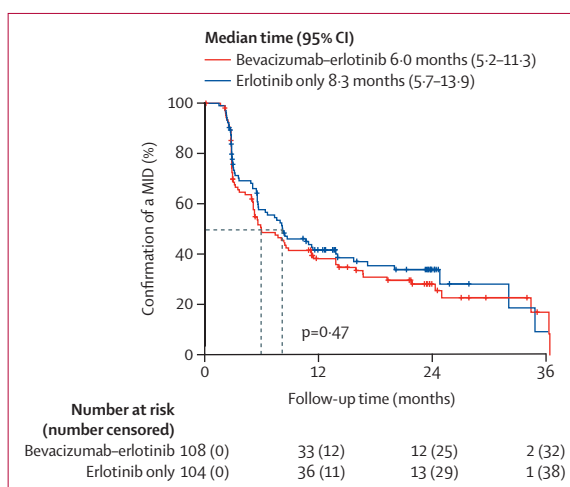


Figure 3: Kaplan-Meier curves for quality of life in the modified intention-to-treat population with completed questionnaires

The tick marks indicate censored observations. The MID was defined as a decline of 1 or more from the total combined score of items 29 and 30 of the European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire-C30, version 3.0. The two-sided p value was calculated by use of a log-rank test. MID=minimally important difference.

QLQ-C30 was 9.1 months (95% CI 6.0–16.8) in the bevacizumab-erlotinib group and 13.9 months (8.3–20.2) in the erlotinib-only group ($p=0.29$; appendix p 12). The median time between enrolment and confirmation of a decline of 3 or more in the total combined score of items 29 and 30 of the EORTC QLQ-C30 was 34.7 months (95% CI 16.8–36.7) in the bevacizumab-erlotinib group and was not reached in the erlotinib-only group ($p=0.68$). Analyses of global health status (quality of life), functional scales, and prespecified patient-reported outcomes (fatigue, pain, and nausea or vomiting) can be found in the appendix (pp 13–21); generally, no differences were found between the groups.

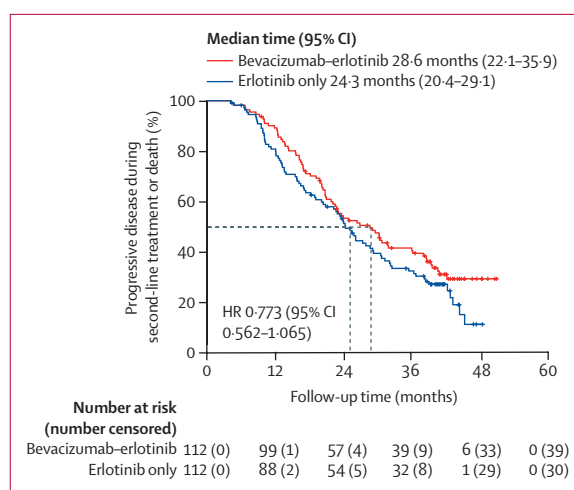


Figure 4: Kaplan-Meier curves for time from enrolment to progressive disease during second-line treatment or death in the modified intention-to-treat population

The tick marks indicate censored observations. HR=hazard ratio.

As for the time from enrolment to progressive disease during second-line treatment or death, after a median follow-up of 23.9 months (IQR 14.2–39.1), there were 73 (65%) events in the bevacizumab-erlotinib group and 82 (73%) events in the erlotinib-only group. The median time from enrolment to progressive disease during second-line treatment or death was slightly longer in the bevacizumab-erlotinib group than in the erlotinib-only group, although the difference was not statistically significant (figure 4).

Discussion

For patients with *EGFR*-mutant NSCLC, the benefit in progression-free survival, the previously reported primary endpoint in the NEJ026 study,¹⁰ produced by adding bevacizumab to erlotinib decreased with time. Here, we show that adding bevacizumab to erlotinib had no significant effect on overall survival or time from enrolment to progressive disease during second-line treatment or death.

We believe that the NEJ026 study better reflected clinical practice than did previous studies (eg, JO25567,⁸ ARCHER 1050,¹⁵ and RELAY¹⁶), in that our study (unlike the others) included patients with an ECOG performance status of 2 or asymptomatic brain metastases. Therefore, our findings could be generalised to diverse medical environments.

There are several possible reasons why the addition of bevacizumab to erlotinib did not impact overall survival. For instance, overall survival might have been influenced by the treatments that were used after disease progression. Owing to the growing number of treatment options for NSCLC, any effect of first-line treatment on overall survival might be skewed by subsequent therapies.^{17,18} Indeed, there has been a progressive increase in post-progression survival over time, probably due to the

deployment of salvage therapy.¹⁹ Thus, improving the overall survival of patients who are already being treated with optimised cancer treatment regimens in Japan might be challenging. Consistent with this hypothesis, the proportions of patients receiving three or more regimens after first-line treatment in the NEJ026 and JO25567 studies⁹ were higher than those in the FLAURA and ARCHER 1050 studies.^{5,15} 30 (27%) patients in the bevacizumab-erlotinib group and 46 (41%) patients in the erlotinib-only group received three or more regimens after the first-line protocol treatment in the NEJ026 study, and 39 (52%) of 75 patients in the bevacizumab-erlotinib group and 42 (55%) of 77 patients in the erlotinib-only group received three or more regimens after the first line in the JO25567 study.⁹ Conversely, 30 (11%) of 279 patients in the osimertinib group and 37 (13%) of 277 patients in the gefitinib-erlotinib group received three or more regimens after the first line in the FLAURA study,⁵ and 27 (12%) of 227 patients in the dacomitinib group and 28 (13%) of 224 patients in the gefitinib group received three or more regimens after the first line in the ARCHER 1050 study.¹⁵ This difference could explain why there was no benefit in overall survival in the bevacizumab-erlotinib group in the NEJ026 and JO25567 studies,⁹ and why overall survival was longer in both of these studies than in the FLAURA⁵ and ARCHER 1050 studies.¹⁵

Patients in the erlotinib only group were more likely to receive more treatments after the protocol regimen than were patients in the bevacizumab-erlotinib group. These tendencies might have been influenced by quality of life, which was improved (non-significantly) in the erlotinib only group compared with the bevacizumab-erlotinib group, could have affected the overall survival result, and were not observed in the JO25567 study.⁹

The RELAY study¹⁶ was a worldwide, double-blind, placebo-controlled, phase 3 study of patients with untreated, *EGFR*-mutant, advanced NSCLC done in 100 hospitals, clinics, and medical centres in 13 countries (including Japan). The trial found that erlotinib plus ramucirumab, a human monoclonal IgG1 VEGFR-2 antagonist, resulted in a longer progression-free survival compared with erlotinib plus placebo (19.4 months [95% CI 15.4–21.6] vs 12.4 months [11.0–13.5]; HR 0.59, 95% CI 0.46–0.76).¹⁶ The overall survival data were immature at the time of data cutoff. Unlike the results of our study, the RELAY study¹⁶ found that the time from enrolment to progressive disease during second-line treatment or death was longer in patients treated with erlotinib-ramucirumab than in patients treated with erlotinib plus placebo (the median was not reached in either group; HR 0.690, 95% CI 0.490–0.972; $p=0.0325$). In our study, platinum plus pemetrexed was recommended for the bevacizumab-erlotinib group and platinum plus pemetrexed plus bevacizumab was recommended for the erlotinib-only group as second-line treatments. Erlotinib alone and erlotinib-bevacizumab, administered as the next treatment after progressive

disease (period 1), were not considered second-line treatments. In the RELAY study,¹⁶ however, any subsequent treatment after discontinuation of the study treatment was at the investigator's discretion; erlotinib could therefore be included if it was considered beneficial to patients. In other words, erlotinib (administered alone after disease progression, as defined by RECIST) was regarded as a second-line treatment in the RELAY study. 61 (27%) of 224 patients in the erlotinib–ramucirumab group and 55 (24%) of 225 patients in the erlotinib-only group received erlotinib alone as a second-line treatment. We found no difference between treatment groups in the exploratory outcome of time from enrolment to progressive disease during second-line treatment or death, and we suggest that this finding reflects the different efficacies of drugs other than erlotinib or bevacizumab. This contrasts with the RELAY study, in which the effects of drugs other than erlotinib cannot easily be determined.

The proportions of patients with Thr790Met *EGFR* mutations at disease progression during first-line treatment were similar between the treatment groups (28% in the bevacizumab–erlotinib group and 26% in the erlotinib-only group). This finding is consistent with a Japanese prospective observational study (REMEDY), in which osimertinib use following resistance to first-generation or second-generation *EGFR* TKIs was approximately 25%.²⁰ Conversely, 11 (15%) of 75 patients in the bevacizumab–erlotinib group and 11 (14%) of 77 patients in the erlotinib-only group received osimertinib in the JO25567 study.⁹ The difference in uptake between our study and the JO25567 study⁹ was probably due to the relative timing of osimertinib approval. The JO25567 study⁹ was done between Feb 21, 2011, and March 5, 2012, whereas the NEJ026 study was done between June 3, 2015, and Aug 31, 2016. On March 28, 2016, osimertinib was approved in Japan for patients with advanced, *EGFR*-mutant, metastatic NSCLC whose disease had progressed on or after first-line *EGFR* TKI therapy and who had developed the secondary Thr790Met mutation.^{21,22} In our study, 43% of patients in the bevacizumab–erlotinib group and 45% of patients in the erlotinib-only group received osimertinib. Osimertinib treatment seemed, post-hoc, to have a small benefit even in later-line settings, and we therefore suggest that re-biopsies and Thr790Met mutation testing are crucial for patients who progress after treatment with first-generation or second-generation *EGFR* TKIs.

There are several limitations to the NEJ026 study. First, the sample size was not sufficient to evaluate whether the treatments afforded a statistically significant benefit to overall survival. When the NEJ026 study protocol was drafted, overall survival data from the JO25567 study were not mature, which precluded us from using the data to calculate the sample size needed for overall survival determination. Furthermore, overall survival is generally longer in Japanese populations compared with other populations,²³ meaning that longer periods and larger

patient cohorts are required to identify whether overall survival has been significantly improved. Therefore, we planned a combined overall survival analysis of the NEJ026 and JO25567 studies as an exploratory endpoint, and hope to report this at a later date.

Second, erlotinib and bevacizumab–erlotinib, used as the next treatment after period 1, were not regarded as second-line treatments. Some patients might benefit from the use of an *EGFR* TKI after progressive disease with first-line treatment.²⁴ Therefore, in clinical practice, *EGFR* TKIs are widely used for progressive disease after first-line treatment. In our study, platinum plus pemetrexed was recommended for the bevacizumab–erlotinib group and platinum plus pemetrexed plus bevacizumab was recommended for the erlotinib-only group as second-line treatments, according to the protocol, differing from clinical practice.

Finally, the NEJ026 study was done only in Japan. Another phase 3 study done in Chinese patients (the CTONG1509 study) found that median progression-free survival after bevacizumab–erlotinib treatment (18.0 months, 95% CI 15.2–20.7) was significantly longer than that after erlotinib treatment (11.3 months, 9.8–13.8; HR 0.55, 95% CI 0.41–0.75; $p < 0.001$).²⁵ However, the overall survival data have not been reported. In addition, the overall survival data of the RELAY study¹⁶ have not been reported because of their immaturity. Therefore, at this time, it is not possible to conclude whether our findings could be generalised outside of Japan.

Osimertinib has now been adopted as the standard first-line treatment for patients with advanced *EGFR*-mutant metastatic NSCLC. Other third-generation *EGFR* TKIs (eg, rociletinib, olmutinib, naquotinib, nazartinib, and PF-06747775) are under development.²⁶ Furthermore, a fourth-generation *EGFR* TKI has also been developed (EAI045).²⁷ The fourth-generation *EGFR* TKIs target the *EGFR* Cys797Ser mutation, which is one of the main mechanisms of resistance to third-generation *EGFR* TKIs.

Although many new *EGFR* TKIs are being developed, combination treatments with *EGFR* TKIs are still attractive options. The NEJ026 study evaluated bevacizumab plus erlotinib, a first-generation *EGFR* TKI. Meanwhile, several studies have already investigated bevacizumab plus osimertinib^{28,29} and other randomised studies are ongoing (eg, UMIN000030206 and NCT04181060). We await these results, and the optimal combination of *EGFR* TKIs and anti-VEGF therapies will be further investigated.

In conclusion, the addition of bevacizumab to erlotinib did not prolong survival, but both treatment groups had relatively long survival durations.

Contributors

MMA was the principal investigator. TF, KH, TN, SM, and KK contributed to the study design, data analysis, and data interpretation. YK, HS, NF, KW, SS, SI, YTsun, OY, MO, KY, IN, MS, KA, FK, YTsuh, YF, HN, GA, SW, and MMi contributed to patient recruitment and data collection. YK and MMA prepared the initial draft of the manuscript, with input from all other authors. All authors approved the final version of the manuscript. YK, SM, KK, and MMA accessed and verified the data.

All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

TF reports personal fees from Chugai Pharmaceutical, during the conduct of the study, and personal fees from AstraZeneca and Boehringer Ingelheim, outside the submitted work. HS reports grants from Chugai Pharmaceutical, AstraZeneca, and MSD, outside the submitted work, and personal fees from Ono Pharmaceutical, Nippon Boehringer Ingelheim, and Novartis Pharma, outside the submitted work. NF reports personal fees from AstraZeneca, Chugai Pharmaceutical, Boehringer Ingelheim, Eli Lilly, Bristol-Myers Squibb, Novartis, Taiho, Kyowa Kirin, MSD, and Pfizer, outside the submitted work. SS reports personal fees from Chugai Pharmaceutical, AstraZeneca, Nippon Boehringer Ingelheim, MSD, Bristol-Myers Squibb, Ono Pharmaceutical, Pfizer, Taiho Pharmaceutical, Eli Lilly, Novartis, Kyowa Hakko Kirin, and Yakult Honsha, outside the submitted work. SI reports grants from Ono Pharmaceutical, and personal fees from Ono Pharmaceutical, Chugai Pharmaceutical, AstraZeneca, MSD, Bristol-Myers Squibb, Eli Lilly Japan, Taiho Pharmaceutical, and Daiichi Sankyo, outside the submitted work. OY reports personal fees from Ono Pharmaceutical, Bristol-Myers Squibb, Chugai Pharmaceutical, AstraZeneca, and Eli Lilly Japan, outside the submitted work. MO reports grants from Chugai Pharmaceutical, during the conduct of the study. MS reports grants and personal fees from Chugai Pharmaceutical, during the conduct of the study. KA reports personal fees from Ono Pharmaceutical, Chugai Pharmaceutical, AstraZeneca, MSD Oncology, and Bristol-Myers Squibb, outside the submitted work. YTs reports personal fees from Chugai Pharmaceuticals, AstraZeneca, and Daiichi Sankyo, outside the submitted work. YF reports personal fees from Eli Lilly, Chugai Pharmaceutical, AstraZeneca, and Bristol-Myers Squibb, outside the submitted work. SW reports grants from AstraZeneca and Boehringer Ingelheim, and personal fees from AstraZeneca, Chugai Pharmaceutical, Ono Pharmaceutical, Bristol-Myers Squibb, Boehringer Ingelheim, Eli Lilly, MSD, Taiho Pharmaceutical, Pfizer, Novartis, and Daiichi Sankyo, outside the submitted work. KH reports grants from Taiho Pharmaceutical, Ono Pharma, and MSD, personal fees from AstraZeneca, Chugai Pharmaceutical, Taiho Pharmaceutical, Ono Pharma, Boehringer Ingelheim, Novartis, Kyorin Pharmaceutical, and Bristol-Myers Squibb, outside the submitted work, and has a patent *EGFR* mutation test licensed to LSI Medience. SM reports personal fees from AstraZeneca, Bristol-Myers Squibb, Chugai Pharmaceutical, Eli Lilly Japan, MSD, Pfizer Japan, Taiho Pharmaceutical, and Ono Pharmaceutical, outside the submitted work. KK reports personal fees from AstraZeneca, Bristol-Myers Squibb Japan, Ono Pharmaceutical, Boehringer Ingelheim, and Taiho Pharmaceutical, outside the submitted work. MMA reports grants and personal fees from Chugai Pharmaceutical during the conduct of the study, and personal fees from Chugai Pharmaceutical, AstraZeneca, Boehringer Ingelheim, and Eli Lilly, outside the submitted work. All other authors declare no competing interests

Data sharing

The data collected for this study will be made available to others in a deidentified form after analyses of all the primary and secondary endpoints have been published, following the completion of a data transfer agreement. Contact the corresponding author at maemondo-ma693@aioros.ocn.ne.jp.

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